Lesson 7 Worksheet: The Challenge of Turning — ANSWER KEY

Part 1: Heading from Coordinates

Current Position	Next Position	row_diff	col_diff	Heading (number and name)
(0, 0)	(1, 0)	1 - 0 = 1	0 - 0 = 0	2 (S)
(2, 3)	(1, 3)	1 - 2 = -1	3 - 3 = 0	0 (N)
(1, 1)	(1, 2)	0	1	1 (E)
(3, 2)	(3, 1)	0	-1	3 (W)
(0, 3)	(1, 3)	1	0	2 (S)
(2, 0)	(2, 1)	0	1	1 (E)
(3, 1)	(2, 1)	-1	0	0 (N)
(0, 2)	(0, 1)	0	-1	3 (W)

Part 2: The Right Turns Table

Current Needed	0 (N)	1 (E)	2 (S)	3 (W)
0 (N)	0	1	2	3
1 (E)	3	0	1	2
2 (S)	2	3	0	1
3 (W)	1	2	3	0

- How many cells have 0 right turns? 4 — Why? When the current heading equals the needed heading, there are zero clockwise steps. These fall along the diagonal.
- What does 2 right turns mean physically? A 180-degree turn (two right turns = turning completely around).

- Do you notice a pattern in the table? Each row is the same sequence [0, 1, 2, 3] shifted. The values only depend on the distance between the headings going clockwise, not the headings themselves.

Part 3: Determine the Number of Right Turns

Scenario	Current Heading	Needed Heading	Clockwise Steps	Right Turns
1	0 (N)	1 (E)	0->1 = 1	1
2	1 (E)	1 (E)	already there = 0	0
3	2 (S)	0 (N)	2->3->0 = 2	2
4	3 (W)	2 (S)	3->0->1->2 = 3	3
5	1 (E)	0 (N)	1->2->3->0 = 3	3
6	0 (N)	3 (W)	0->1->2->3 = 3	3
7	2 (S)	3 (W)	2->3 = 1	1
8	3 (W)	1 (E)	3->0->1 = 2	2

- Why does counting clockwise always give the correct number of right turns? Each clockwise step corresponds to one right turn. Since the headings are arranged in a circle (0->1->2->3->0), counting steps clockwise from current to needed always gives the number of right turns needed. For example, going from 3 (W) to 1 (E): count 3->0->1, that is 2 steps, so 2 right turns. This works even when you have to "wrap around" past 0.

Part 4: Trace a Path with Heading Updates

Path: [(1, 0), (2, 0), (2, 1), (2, 2), (2, 3)], Starting position: (0, 0), Starting heading: 0 (N)

Step	Position	Next	Needed Heading	Current Heading	Right Turns	New Heading
1	(0, 0)	(1, 0)	2 (S)	0 (N)	count 0->1->2 = 2	2 (S)
2	(1, 0)	(2, 0)	2 (S)	2 (S)	already there = 0	2 (S)
3	(2, 0)	(2, 1)	1 (E)	2 (S)	count 2->3->0->1 = 3	1 (E)
4	(2, 1)	(2, 2)	1 (E)	1 (E)	already there = 0	1 (E)

Step	Position	Next	Needed Heading	Current Heading	Right Turns	New Heading
5	(2, 2)	(2, 3)	1 (E)	1 (E)	already there = 0	1 (E)

- How many steps required turning? 2 (step 1 with 2 right turns and step 3 with 3 right turns)
- How many steps required zero turns? 3 (steps 2, 4, 5)
- At which step did the robot change from row to column movement? Step 3

Part 5: Trace a Second Path

Path: [(1, 3), (0, 3), (0, 2), (0, 1), (0, 0)], Starting position: (2, 3), Starting heading: 2 (S)

Step	Position	Next	Needed Heading	Current Heading	Right Turns	New Heading
1	(2, 3)	(1, 3)	0 (N)	2 (S)	count 2->3->0 = 2	0 (N)
2	(1, 3)	(0, 3)	0 (N)	0 (N)	already there = 0	0 (N)
3	(0, 3)	(0, 2)	3 (W)	0 (N)	count 0->1->2->3 = 3	3 (W)
4	(0, 2)	(0, 1)	3 (W)	3 (W)	already there = 0	3 (W)
5	(0, 1)	(0, 0)	3 (W)	3 (W)	already there = 0	3 (W)

Final heading after completing path: 3 (W)

Part 6: Trace a Third Path

Path: [(1, 2), (1, 3), (2, 3), (3, 3)], Starting position: (1, 1), Starting heading: 0 (N)

Step	Position	Next	Needed Heading	Current Heading	Right Turns	New Heading
1	(1, 1)	(1, 2)	1 (E)	0 (N)	count 0->1 = 1	1 (E)

Step	Position	Next	Needed Heading	Current Heading	Right Turns	New Heading
2	(1, 2)	(1, 3)	1 (E)	1 (E)	already there = 0	1 (E)
3	(1, 3)	(2, 3)	2 (S)	1 (E)	count 1->2 = 1	2 (S)
4	(2, 3)	(3, 3)	2 (S)	2 (S)	already there = 0	2 (S)

- Final heading: **2 (S)**
- Does the first step require turning? **Yes** — Why? **The robot starts facing 0 (North) but needs heading 1 (East), so it must make 1 right turn: count clockwise 0->1 = 1 step.**

Part 7: Robot Orientation Diagrams

- Scenario A: Facing 0 (N), need heading 1 (E) → Right turns needed: **1** — count 0->1 = 1
- Scenario B: Facing 1 (E), need heading 2 (S) → Right turns needed: **1** — count 1->2 = 1
- Scenario C: Facing 3 (W), need heading 1 (E) → Right turns needed: **2** — count 3->0->1 = 2
- Scenario D: Facing 2 (S), need heading 2 (S) → Right turns needed: **0** — already there

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Part 8: Heading Logic in Code

```
HEADING_NAMES = ["N", "E", "S", "W"]

def get_needed_heading(current_pos, next_pos):
    row_diff = next_pos[0] - current_pos[0]
    col_diff = next_pos[1] - current_pos[1]

    if row_diff == -1:
        return 0      # North
    elif col_diff == 1:
        return 1      # East
    elif row_diff == 1:
        return 2      # South
    elif col_diff == -1:
        return 3      # West
```

Test your understanding:

1. `get_needed_heading((0,0), (1,0))` returns **2** — row_diff = 1, so heading is 2 (South)
2. `get_needed_heading((2,3), (2,2))` returns **3** — col_diff = -1, so heading is 3 (West)
3. `get_needed_heading((1,1), (0,1))` returns **0** — row_diff = -1, so heading is 0 (North)

Reflection

What makes numeric headings and clockwise counting so powerful for solving the turning problem? Using numbers 0-3 arranged clockwise means we can count steps from any heading to any other heading using a simple while loop. The loop walks clockwise one step at a time (adding 1 and wrapping from 3 back to 0), counting as it goes. This works for all 16 combinations automatically. It directly mirrors what the robot does physically – each step in the count is one right turn. The approach is easy to understand, easy to code, and reveals the underlying structure: headings are arranged in a circle, and counting clockwise naturally wraps around.